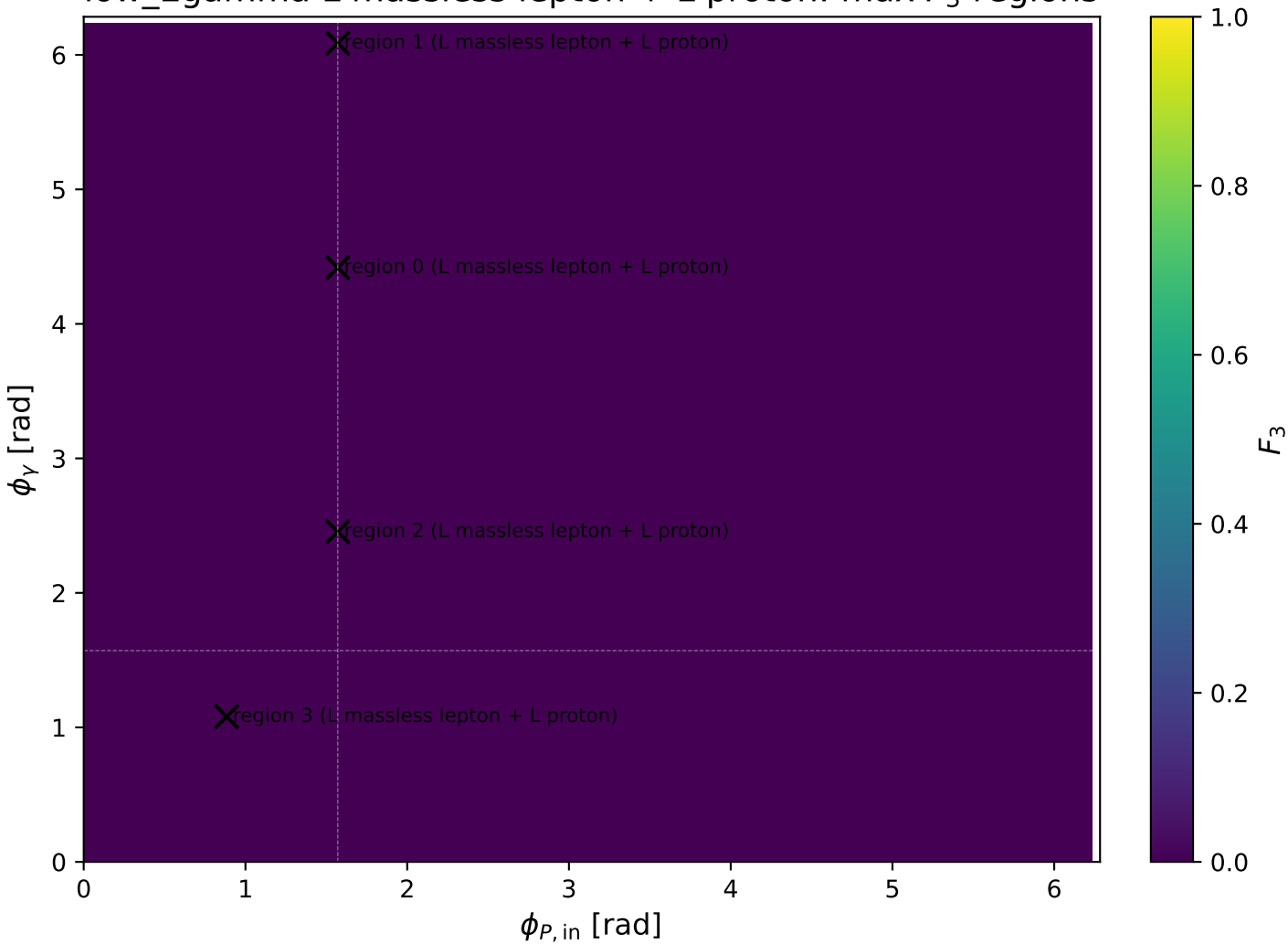
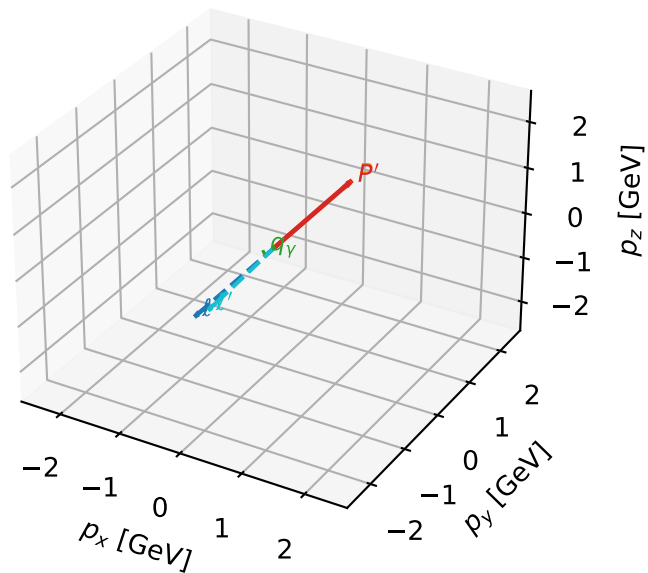


low_Egamma L massless lepton + L proton: max F_3 regions



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

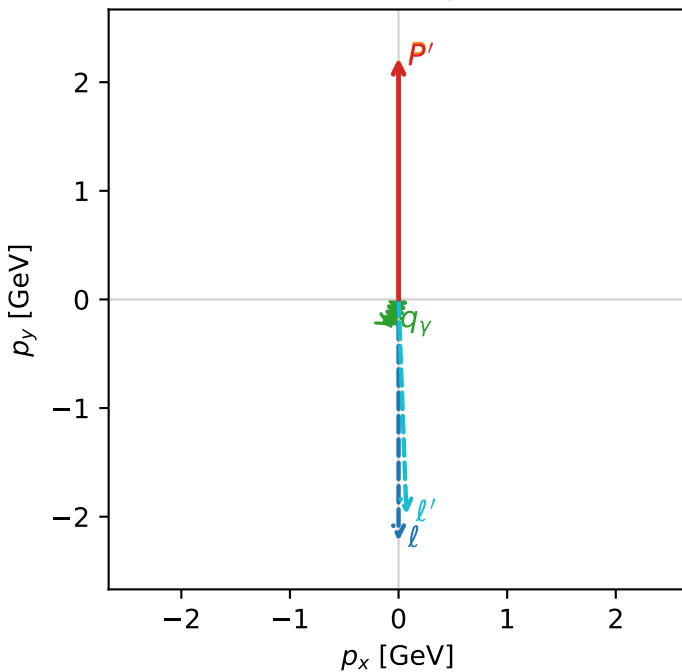


f3_low_egamma_ll_region_0 F_3=2.38883e-08
 region: low_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s high_theta_in_low_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.21812$
 $\theta_{in}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_p=1.5708$
 $E_Y=0.25$, $\phi_Y=4.41786$
 $Q^2=0.00591804$, $x_B=0.00260549$, $t=-5.99798e-06$, $W^2=3.1453$, $y=0.110068$
 $\theta(\ell', \gamma)=18.9753$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.9802$ $p_x= 0.0726$ $p_y= -1.9789$ $p_z= 0.0000$
 P' $E= 2.4083$ $p_x= 0.0000$ $p_y= 2.2181$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.0726$ $p_y= -0.2392$ $p_z= 0.0000$

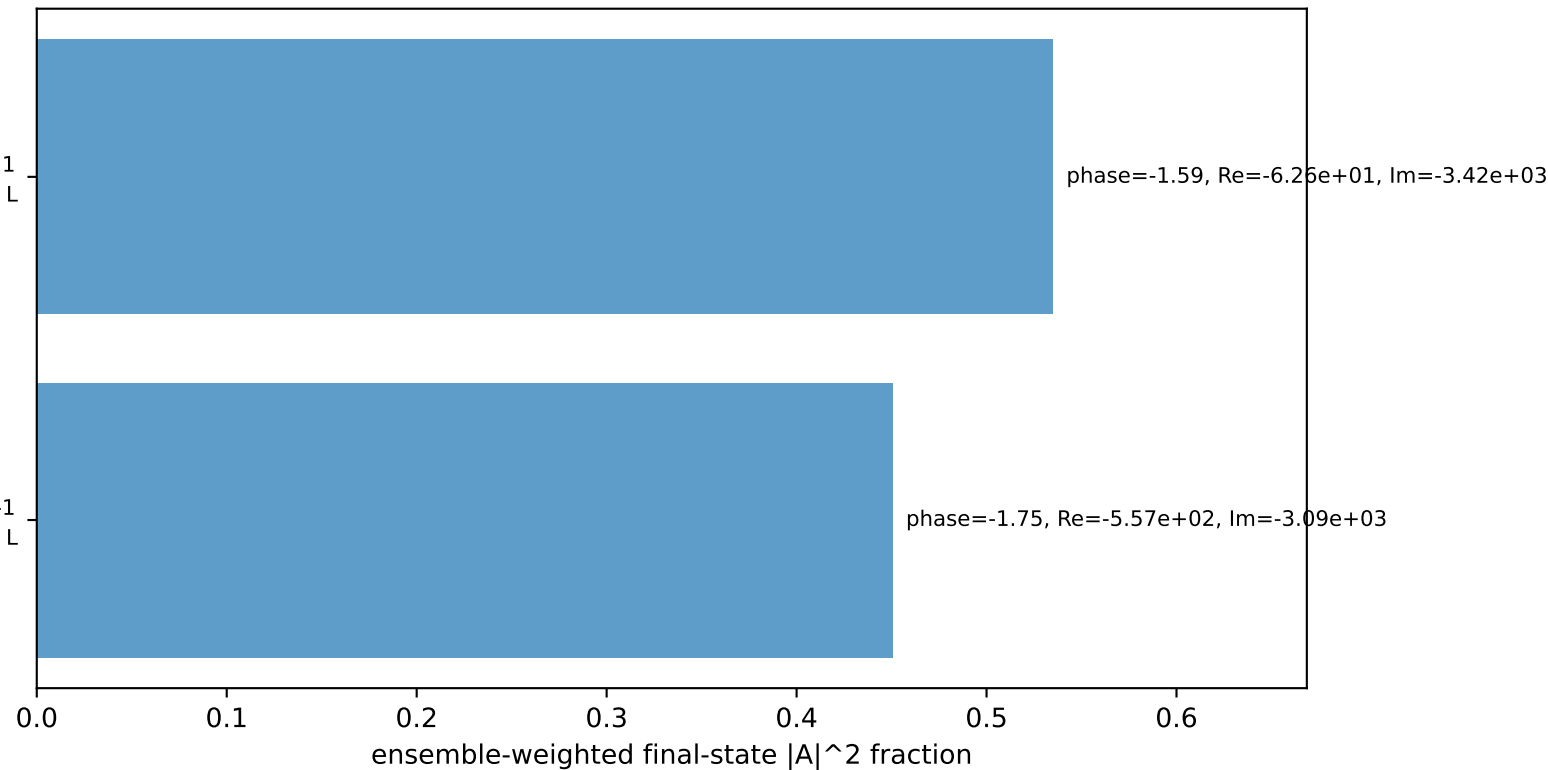
Transverse plane



$h_e=+1, h_p=-1, h_Y=+1$
 massless lepton L, proton L

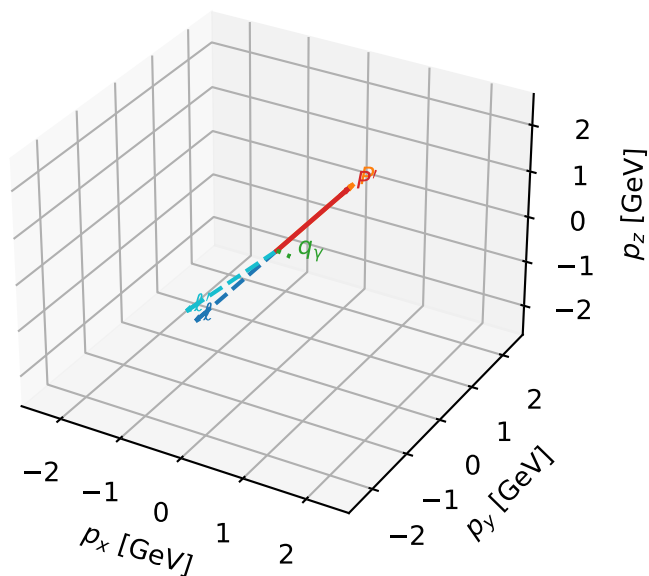
$h_e=+1, h_p=+1, h_Y=-1$
 massless lepton L, proton L

Leading initial-ensemble/final-state components (N=2, retained=98.5%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

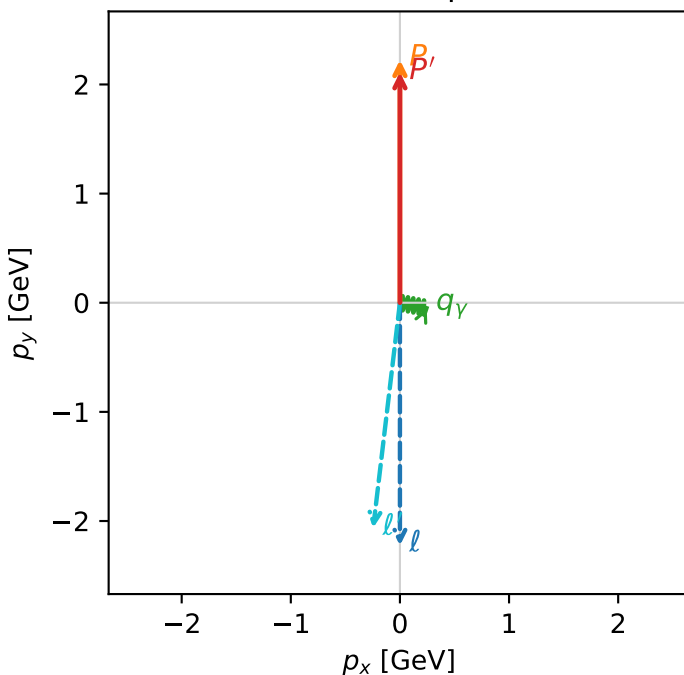


f3_low_egamma_ll_region_1 F_3=5.73407e-09
 region: low_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.11192$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_P=1.5708$
 $E_Y=0.25$, $\phi_Y=6.08684$
 $Q^2=0.0645935$, $x_B=0.0452961$, $t=-0.00199627$, $W^2=2.24128$, $y=0.0691039$
 $\theta(\ell', \gamma)=85.5276$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 2.0777$ $p_x= -0.2452$ $p_y= -2.0631$ $p_z= 0.0000$
 P' $E= 2.3109$ $p_x= 0.0000$ $p_y= 2.1119$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.2452$ $p_y= -0.0488$ $p_z= 0.0000$

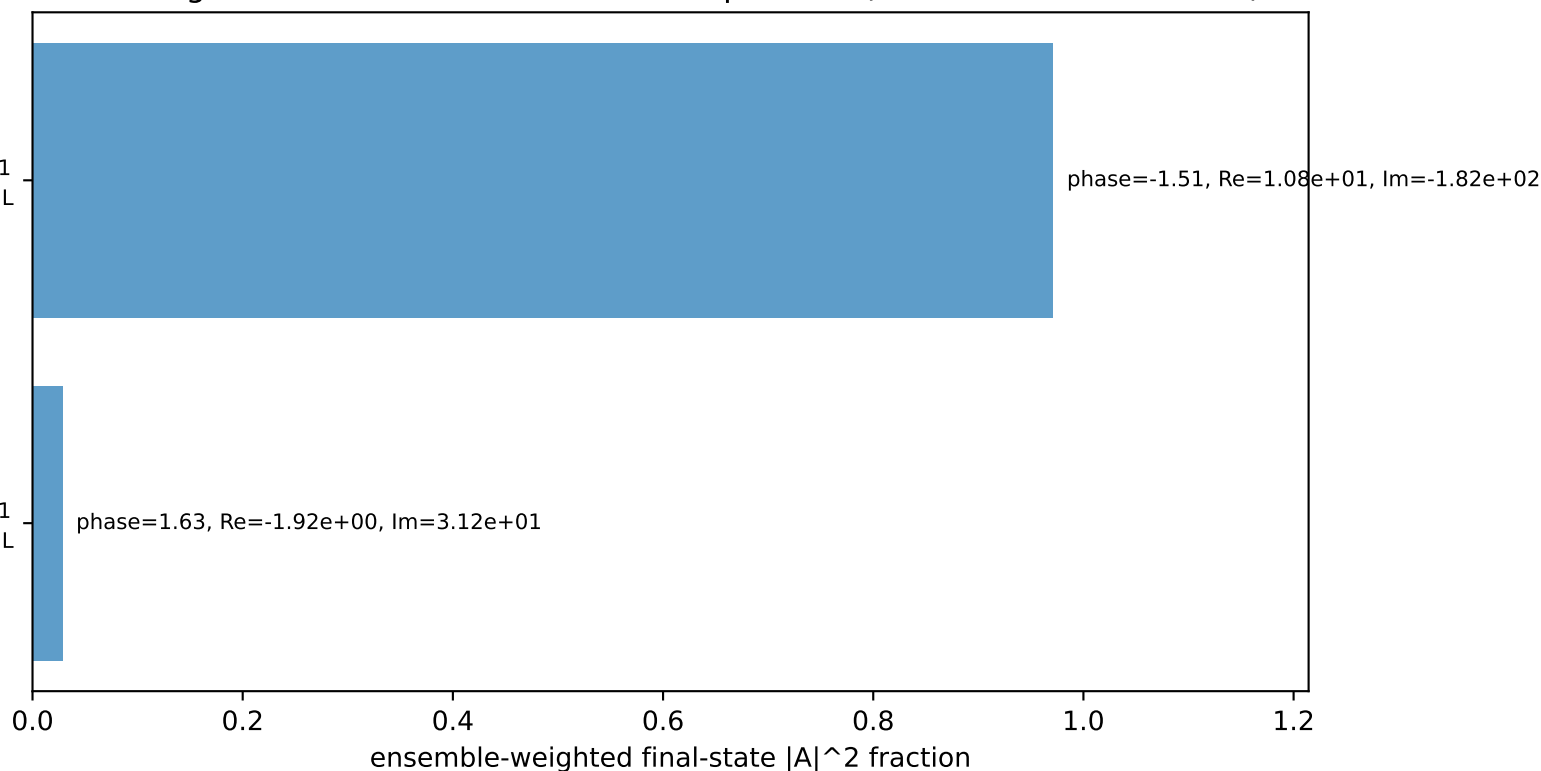
Transverse plane



$h_e=+1, h_p=-1, h_Y=+1$
 massless lepton L, proton L

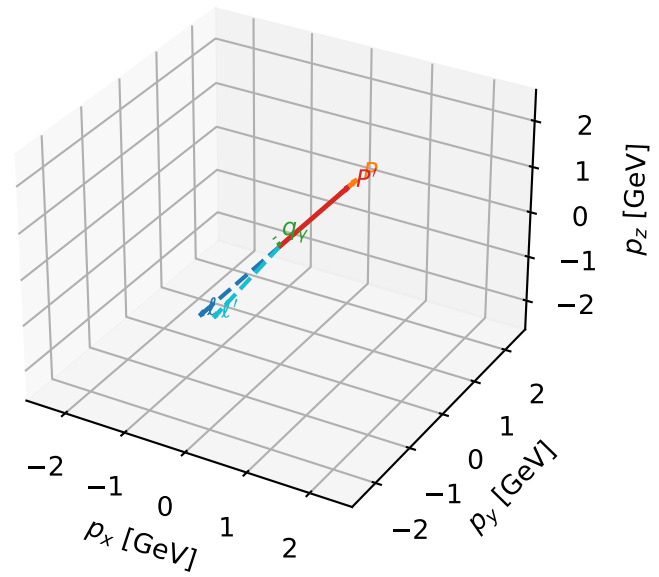
$h_e=+1, h_p=+1, h_Y=-1$
 massless lepton L, proton L

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta



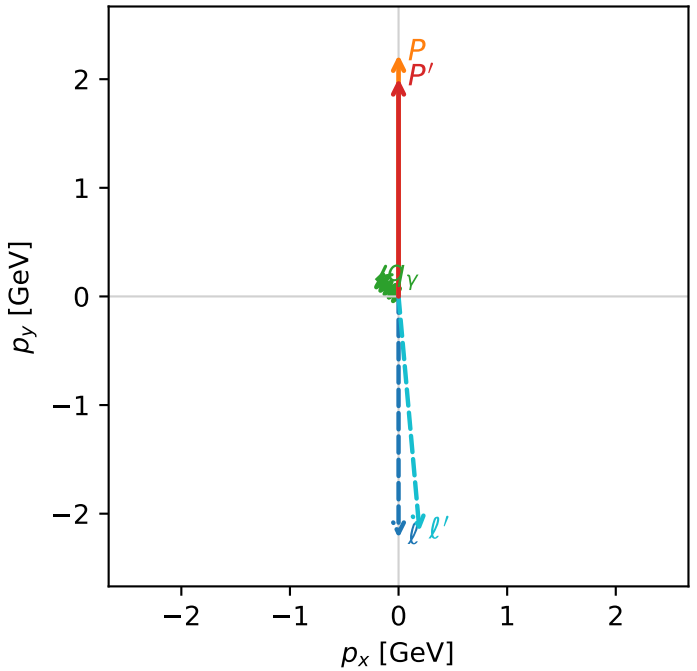
f3_low_egamma_ll_region_2 F_3=2.39171e-09
region: low_Egamma_LL_region_2
outgoing-state purity: 1
kinematic point: high_s high_theta_in_low_Egamma
incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.00644$
 $\theta_{in}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_p=1.5708$
 $E_{\gamma}=0.25$, $\phi_{\gamma}=2.45437$
 $Q^2=0.0382947$, $x_B=0.0751918$, $t=-0.00782133$, $W^2=1.35084$, $y=0.0246799$
 $\theta(\ell', \gamma)=134.476$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

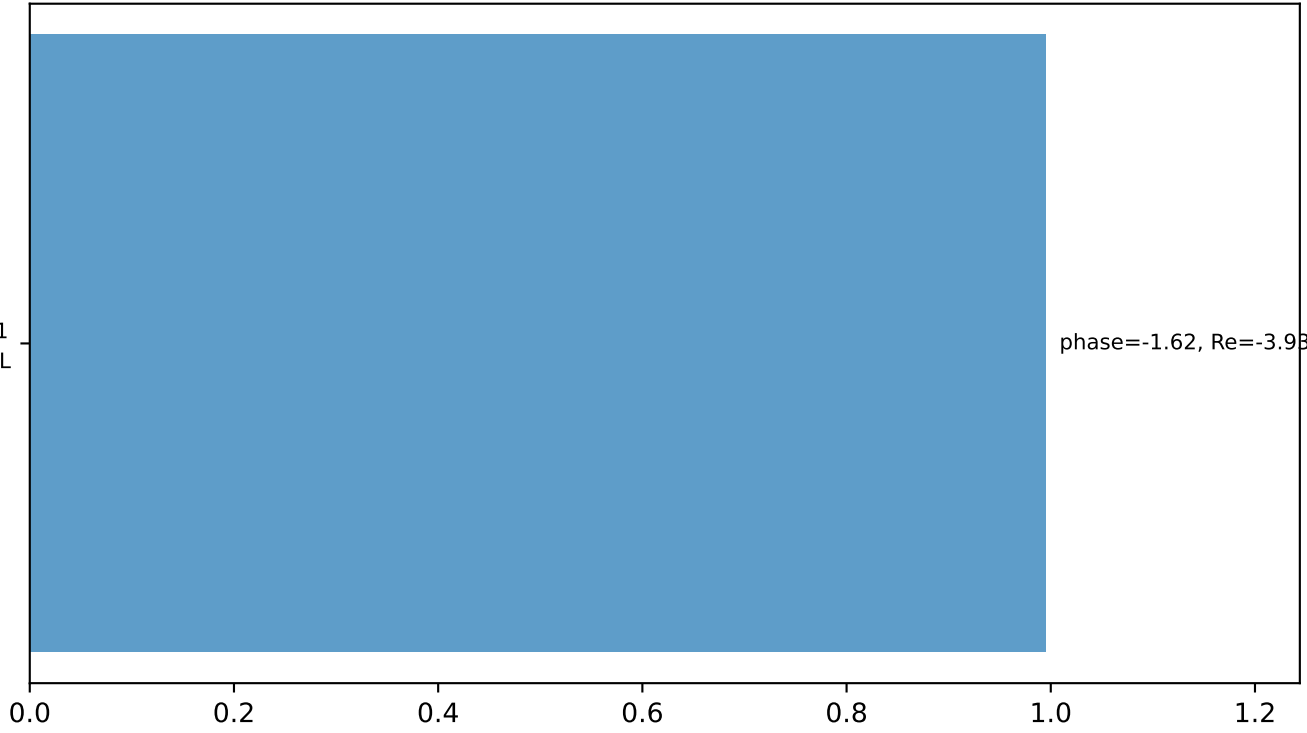
ℓ	$E=$	2.2244	$p_x=$	-0.0000	$p_y=$	-2.2244	$p_z=$	-0.0000
P	$E=$	2.4141	$p_x=$	0.0000	$p_y=$	2.2244	$p_z=$	0.0000
ℓ'	$E=$	2.1736	$p_x=$	0.1933	$p_y=$	-2.1650	$p_z=$	0.0000
P'	$E=$	2.2149	$p_x=$	0.0000	$p_y=$	2.0064	$p_z=$	0.0000
q_{γ}	$E=$	0.2500	$p_x=$	-0.1933	$p_y=$	0.1586	$p_z=$	0.0000

Transverse plane



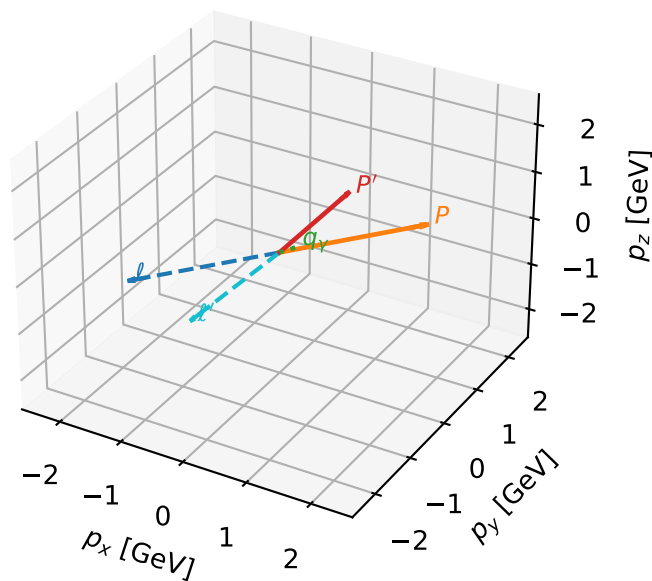
$h_e=+1, h_p=-1, h_{\gamma}=+1$
massless lepton L, proton L

Leading initial-ensemble/final-state components (N=1, retained=99.5%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

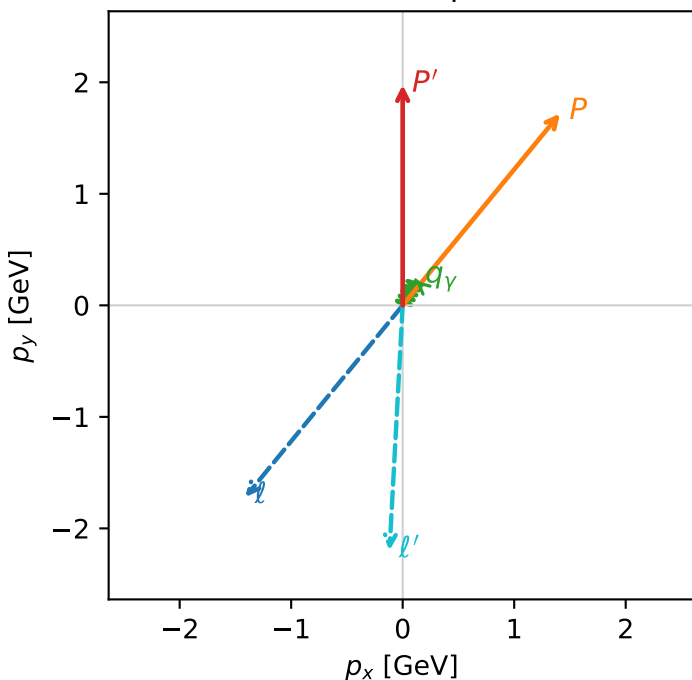


f3_low_egamma_ll_region_3 F_3=1.52492e-09
 region: low_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_{l_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.97681$
 $\theta_{in}=1.57079$, $\phi_l=4.02517$, $\phi_p=0.883573$
 $E_\gamma=0.25$, $\phi_\gamma=1.07992$
 $Q^2=1.90036$, $x_B=0.895257$, $t=-2.00648$, $W^2=1.10218$, $y=0.102864$
 $\theta(l', \gamma)=154.945$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -1.4112$ $p_y= -1.7195$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 1.4112$ $p_y= 1.7195$ $p_z= 0.0000$
 l' $E= 2.2005$ $p_x= -0.1178$ $p_y= -2.1973$ $p_z= 0.0000$
 P' $E= 2.1881$ $p_x= 0.0000$ $p_y= 1.9768$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.1178$ $p_y= 0.2205$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)

